

Who-Speak-AI: Secure Virtual Assistant with Speaker Recognition

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I Introduction

1. Background

Voice-based virtual assistants have advanced rapidly alongside broader progress in Artificial Intelligence, evolving from simple command-response systems into sophisticated conversational agents capable of complex reasoning and task execution. Despite their growing capabilities, widespread adoption has surfaced critical concerns regarding user privacy and security.

In particular, sensitive user information such as behavioral patterns, personal preferences, and private data must be safeguarded against unauthorized access. Reliably verifying a user's identity before processing such information is therefore an essential prerequisite for building trustworthy voice-based assistants. Speaker identification, the task of determining *who* is speaking from an audio signal, offers a natural and non-intrusive means of achieving this verification.

Beyond security, robust speaker identification also enables richer, more personalized user experiences. By recognizing individual speakers, a virtual assistant can tailor its responses, recommendations, and guidance to each user's unique context and history.

However, speaker identification in real-world settings remains a challenging task. Spoken utterances vary considerably due to factors such as background noise, recording conditions, emotional state, and dialectal differences. Moreover, models trained on data from one linguistic or acoustic domain may generalize poorly to others, underscoring the need for evaluation across diverse and representative datasets. These challenges motivate the adoption of modern deep learning architectures that can learn robust speaker representations directly from raw or lightly processed audio.

Motivated by these considerations, this project provides a concise introduction to speaker identification and presents a systematic analysis and comparison of several methods and models grounded in Statistical Machine Learning, with the overarching goal of advancing speaker identification for reliable and efficient virtual assistants.

2. Our Contributions

The main contributions of this project are threefold:

1. **Dataset curation.** We select two publicly available datasets tailored to our concerns regarding practical speaker identification. The *ViMD* dataset captures diverse dialectal variation across Vietnamese regions, while the *TidyVoice2026* dataset provides a broader, multilingual corpus that tests robustness for global deployment. Together, these datasets allow us to evaluate model performance under both language-specific and language-agnostic conditions.
2. **Model benchmarking.** We systematically benchmark two state-of-the-art speaker identification models—RawNet3 and ECAPA-TDNN—on both datasets, analyzing their strengths

and limitations to draw actionable conclusions for real-world use. Our evaluation covers multiple metrics and investigates how each architecture responds to domain shift between the two corpora.

3. **Proof-of-concept application.** We develop *Who-Speak-AI*, a working voice-based virtual assistant that demonstrates secure, speaker-aware interaction. The application adopts a modular pipeline comprising Voice Activity Detection (VAD), Automatic Speech Recognition (ASR), a Large Language Model (LLM) for dialogue, and Text-to-Speech (TTS) synthesis. This end-to-end prototype illustrates how speaker identification can be seamlessly integrated into a practical conversational system.

3. Report Organization

The remainder of this report is organized as follows. **Section II** reviews related works on speaker identification and recognition. **Section III** describes the two datasets used in our experiments. **Section IV** presents the architectures of the models under study. **Section V** details the experimental setup and evaluation methodology. **Section VI** reports and analyzes the experimental results. **Section VII** introduces the Who-Speak-AI application as a proof-of-concept deployment. Finally, **Section VIII** concludes the report and discusses future directions.

II Related Works

This section surveys the key developments in speaker recognition that inform our work, organized into three areas: the evolution of speaker embedding architectures, advances in speaker enrollment strategies, and the emerging role of phonetic information in speaker modeling.

1. Speaker Embedding Architectures

Early speaker recognition systems represented speakers using statistical models of hand-crafted acoustic features. The most influential of these was the **i-vector** framework [1], which employed a Gaussian Mixture Model–Universal Background Model (GMM-UBM) pipeline to compress variable-length utterances into fixed-dimensional vectors through factor analysis. Although i-vectors became a de facto standard for over a decade, their reliance on shallow generative modeling left them sensitive to channel mismatch, background noise, and speaking-style variation.

Deep neural networks have since supplanted statistical approaches. Snyder et al. introduced **x-vectors** [2], replacing the GMM back-end with a discriminatively trained neural architecture. The x-vector system feeds MFCC features through a stack of **Time Delay Neural Network (TDNN)** layers [3] that model temporal context at the frame level, aggregates them via a statistics pooling layer, and classifies speakers with a softmax objective. By learning representations end-to-end, x-vectors substantially outperformed i-vectors across multiple benchmarks.

Building on the TDNN backbone, Desplanques et al. proposed **ECAPA-TDNN** [4], which augments the x-vector paradigm with Squeeze-and-Excitation blocks for channel-wise attention, multi-scale feature aggregation, and attentive statistics pooling. Dawalatabad et al. [5] subsequently demonstrated that ECAPA-TDNN embeddings generalize well beyond verification to speaker diarization tasks. Training with the **ArcFace** angular-margin loss [6] further sharpens inter-speaker discrimination, yielding an Equal Error Rate (EER) of 0.87% on the **VoxCeleb1** test set [7]. Meanwhile, Jung et al. pursued a complementary direction with the **RawNet** family [8, 9], culminating in **RawNet3** [10], which operates directly on raw waveforms and eliminates the need for hand-designed spectral features altogether. These two architectures—ECAPA-TDNN and RawNet3—represent contrasting design philosophies (engineered features vs. learned front-ends) and form the basis of our comparative study.

2. Speaker Enrollment Strategies

Reliable speaker identification depends not only on the embedding model but also on the quality of the enrollment data used to construct each speaker’s reference profile. Several lines of work have sought to improve enrollment.

Li et al. [11] proposed **enrollment speech augmentation**, applying noise and reverberation perturbations to enrollment utterances so that the resulting speaker template becomes

more resilient to acoustic mismatch at test time. Mingote et al. [12] took a model-centric approach with **network-optimized enrollment models**: instead of averaging embeddings, they trained a per-speaker output head whose weights serve as the speaker template, eliminating the need for a separate scoring back-end. Le and Ngo [13] introduced a **data-centric enrollment selection** algorithm that iteratively retains only those utterances whose pairwise cosine similarities exceed an adaptive threshold, filtering out noisy or atypical recordings while preserving representative ones. Despite their effectiveness, all of these methods operate on the acoustic and environmental dimension of enrollment quality, largely ignoring the linguistic content of the enrolled speech.

3. Phonetic Information in Speaker Modeling

A parallel body of research has established that phonetic content plays a significant role in how well speaker characteristics are captured. Liu et al. [14, 15] conducted early investigations showing that **phoneme-aware speaker embeddings**—obtained by conditioning the network on phonetic labels—consistently improve verification accuracy over content-agnostic baselines. Wang et al. [16] corroborated this finding for text-independent scenarios, demonstrating that even implicit phonetic structure benefits embedding quality. Subsequent work has refined these ideas with more sophisticated architectures: Liu et al. [17] introduced a **phonetic attention mask** that guides the network to weight phonetically informative frames more heavily, while Hong et al. [18] proposed **phonetic decomposition and reorganization** modules that disentangle speaker and phonetic factors within the embedding space. Most recently, Song et al. [19] extended these techniques to the **multilingual** setting, showing that cross-lingual phonetic features transfer effectively for speaker verification across languages. Complementing these neural approaches, Dellwo et al. [20] provided acoustic-phonetic evidence that **prosodic and rhythmic variability** across a wider phonetic range better captures individual speaker traits.

In the enrollment domain, the **TIMIT** corpus [21]—a phonetically balanced speech database originally designed for ASR research—illustrates the long-recognized principle that controlled phonetic coverage leads to more informative speech samples. Yet, to the best of our knowledge, no prior work has explicitly leveraged phonetically rich, guided utterances during the enrollment phase for speaker identification. Le and Ngo [22] recently addressed this gap by proposing a **phoneme-based enrollment optimization** strategy that selects enrollment sentences to maximize phoneme coverage, thereby producing higher-quality speaker profiles. Their results demonstrate that bridging linguistic diversity with enrollment design leads to more robust and consistent recognition, and their approach provides a key foundation for the enrollment methodology adopted in this project.

III Datasets

While VoxCeleb [7] and VoxCeleb2 [23] remain the de facto benchmarks for speaker recognition, they are unsuitable for our benchmark for three key reasons. First, both candidate models were already pretrained on VoxCeleb; evaluating on the same corpus would confound downstream validation with pre-training exposure and fail to assess target-domain adaptation. Second, VoxCeleb’s English-centric broadcast speech cannot evaluate robustness to *dialectal variation* across Vietnamese regions. Third, it lacks the typological breadth needed to test multilingual generalization. We therefore adopt two complementary datasets: the **ViMD** corpus [24] for fine-grained Vietnamese dialect coverage, and the **TidyVoice** corpus [25] for large-scale multilingual evaluation.

1. ViMD

The Vietnamese Multi-Dialect dataset (ViMD) [24], introduced at EMNLP 2024, contains approximately 102.56 hours of speech across roughly 19,000 utterances spanning all 63 provinces of Vietnam. Recordings are annotated with speaker identity codes (e.g., `spk_{province}_{id}`), gender labels, and province-level dialect tags. ViMD is valuable for our study because speakers from different provinces exhibit systematic phonetic and prosodic differences that a naïve model might conflate with speaker identity, requiring models to disentangle vocal characteristics from dialectal variation.

Table 1 summarizes the key statistics of the ViMD dataset as used in our experiments.

Property	Value
Total duration (hours)	≈102.56
Total utterances (processed)	18,947 (Train: 15,023, Val: 1,898, Test: 2,026)
Total speakers (processed)	12,953 (Train: 10,291, Val: 1,318, Test: 1,344)
Provinces / dialects	63 provinces
Sample format	16 kHz mono (resampled from 44.1/48 kHz)
Speaker leakage	0 (pruned overlapping singleton speakers)

Table 1: Summary statistics of the ViMD dataset splits.

2. TidyVoice

The TidyVoice dataset [25] is a multilingual corpus curated from Mozilla Common Voice with a rigorous speaker-identity cleaning pipeline that resolves the label noise inherent in raw Common Voice client IDs. It comprises two partitions:

- **Tidy-M** (monolingual): over 212,000 speakers across 81 languages.

- **Tidy-X** (cross-lingual): approximately 4,500 multilingual speakers across 40 languages, designed for cross-lingual speaker verification.

TidyVoice also underpins the **TidyVoice 2026 Challenge** at Interspeech 2026. It complements ViMD by testing whether speaker embeddings generalize across entirely different languages and recording environments—critical for globally deployed virtual assistants.

Table 2 presents the key statistics of the TidyVoice partition used in our experiments.

Property	Value
Source corpus	Mozilla Common Voice (Tidy-M partition)
Total utterances	321,711 (Train: 262,268, Val: 29,720, Test: 29,723)
Total speakers	4,474 (Train: 3,666, Val: 404, Test: 404)
Speaker overlap	0 (disjoint speaker sets across splits)
Sample format	16 kHz mono PCM
Split balancing	Language distribution & utterance count

Table 2: Summary statistics of the TidyVoice benchmark partition.

IV Model

This section describes the two speaker embedding architectures evaluated in our study. Both produce fixed-dimensional embeddings from variable-length audio but differ in their front-end design: ECAPA-TDNN operates on handcrafted Mel-frequency features, while RawNet3 learns directly from raw waveforms.

1. ECAPA-TDNN

ECAPA-TDNN, proposed by Desplanques et al. [4], enhances the x-vector TDNN architecture [2] with three innovations: SE-Res2Blocks, multi-layer feature aggregation, and channel-dependent attentive statistics pooling.

1.1 SE-Res2Block

The core building block (Figure 1) is a bottleneck residual unit. Two pointwise Conv1D layers sandwich a **Res2 dilated convolution**—a module from Res2Net that splits channels into $s = 8$ groups and processes them hierarchically, enabling multi-scale temporal modeling within a single layer. A **Squeeze-and-Excitation (SE)** module then recalibrates channels by computing global temporal statistics and learning per-channel scaling weights, effectively injecting utterance-level context into frame-level features. A skip connection completes the block.

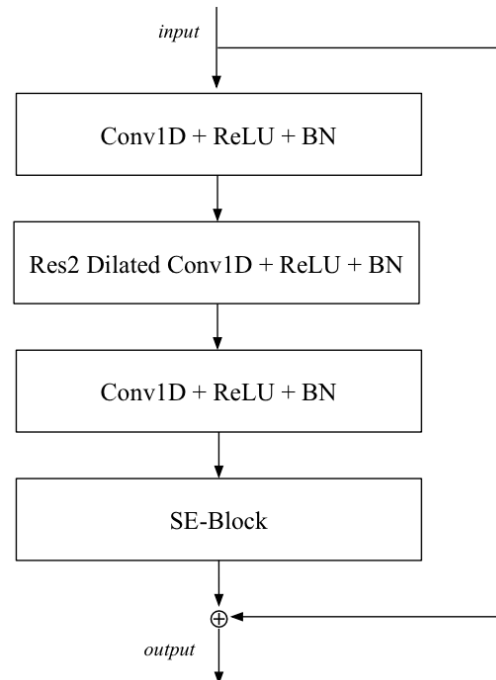


Figure 1: SE-Res2Block of ECAPA-TDNN [4].

1.2 Full Architecture

As shown in Figure 2, 80-dim MFCCs pass through an initial Conv1D ($k = 5$), then three SE-Res2Blocks with increasing dilation ($d = 2, 3, 4$). The residual connection for each block is

defined as the sum of *all* preceding block outputs, not just the immediate input.

The outputs of all three blocks are concatenated (**multi-layer feature aggregation**) and fused by a Conv1D ($k = 1$) into $1536 \times T$ features. A **channel-dependent attentive statistics pooling** layer computes per-channel attention weights—augmented with a global context vector (utterance-level mean and standard deviation)—to produce weighted means and standard deviations (3072×1). A final FC layer yields a 192-dim embedding, trained with AAM-Softmax [6].

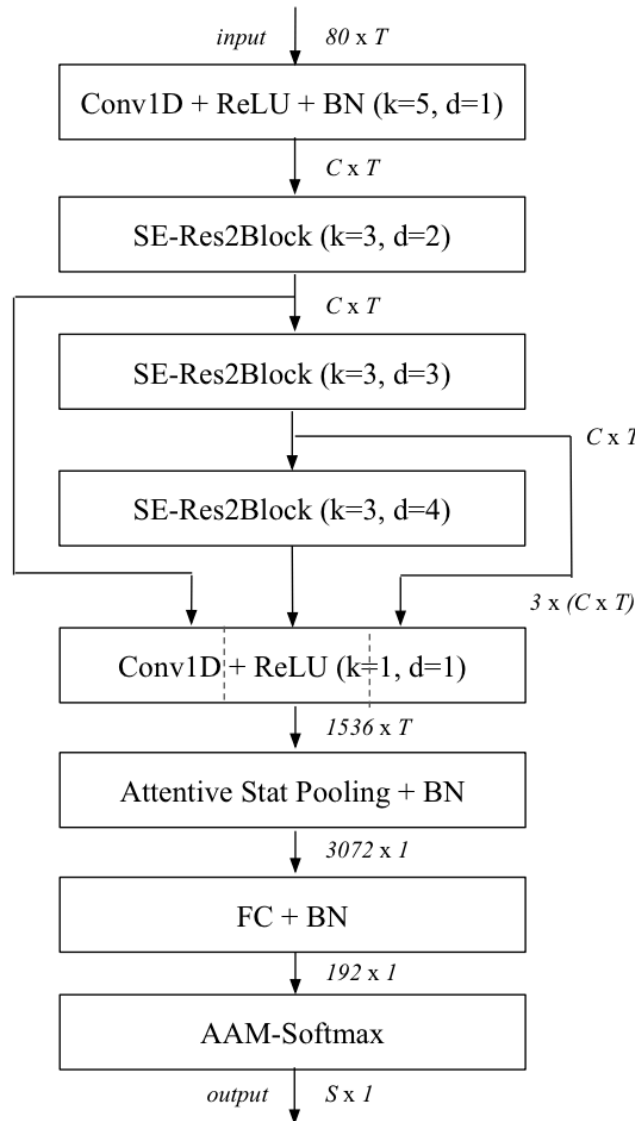


Figure 2: Full ECAPA-TDNN architecture [4].

2. RawNet3

RawNet3, proposed by Jung et al. [10], is the third generation of the RawNet family [8, 9]. It adopts ECAPA-TDNN’s macro-topology but replaces the handcrafted front-end with a learnable one and adapts the building blocks for raw waveform input.

2.1 AFMS-Res2MP-Block

The backbone block (Figure 3) mirrors the SE-Res2Block with two modifications. First, **max pooling** is applied after the residual addition to reduce the much longer raw waveform sequences (pool sizes $p = 5, 3, 1$ across blocks). Second, **Adaptive Feature Map Scaling (AFMS)**—inherited from RawNet2—replaces the SE module, applying a learned affine channel-wise transformation based on global statistics.

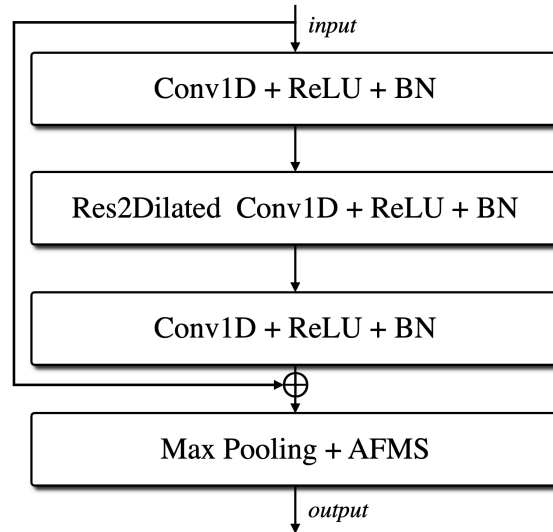


Figure 3: AFMS-Res2MP-block of RawNet3 [10].

2.2 Full Architecture

As shown in Figure 4, raw waveforms ($1 \times T$) first pass through pre-emphasis, instance normalization, and a **parameterized analytic filterbank**—a learnable Conv1D ($k=251$, stride S) that replaces fixed MFCC extraction, followed by log compression and mean normalization.

Three AFMS-Res2MP-blocks with parameters $(k, d, p) = (3, 2, 5), (3, 3, 3), (3, 3, 1)$ process the output. Following ECAPA-TDNN, all block outputs are concatenated and fused by a Conv1D. **Channel and context statistics pooling** (replacing RawNet2’s GRU) collapses the temporal axis into 3072×1 . A FC layer produces a 256-dim embedding, trained with AAM-Softmax.

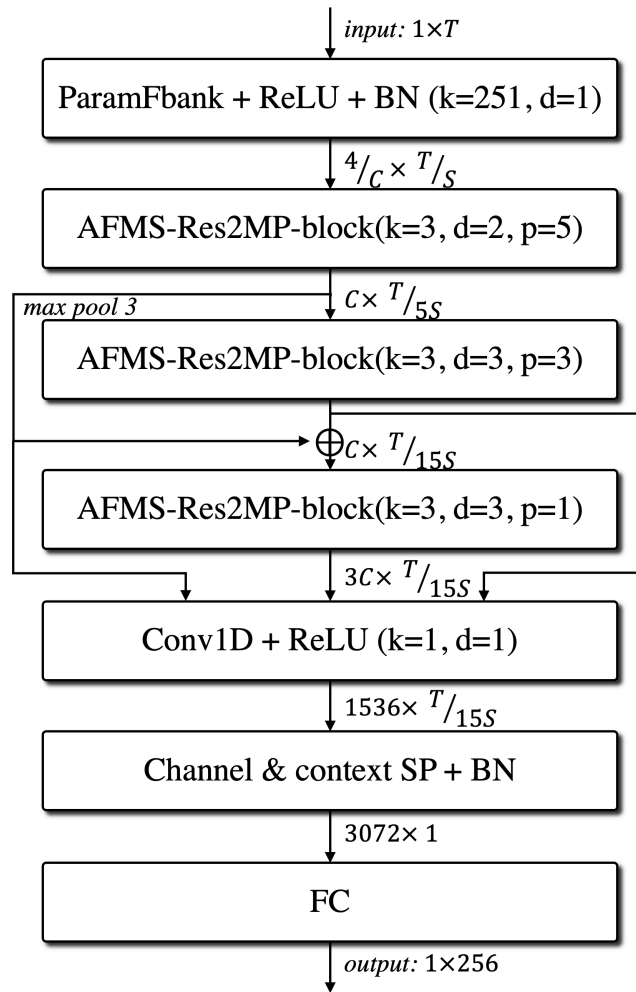


Figure 4: Full RawNet3 architecture [10].

3. Architectural Comparison

Property	ECAPA-TDNN	RawNet3
Input	80-dim MFCC	Raw waveform
Front-end	Fixed (handcrafted)	Learned (analytic fbank)
Core block	SE-Res2Block	AFMS-Res2MP-block
Channel attention	Squeeze-Excitation	AFMS
Sequence reduction	—	Max pooling
Pooling	Channel-dep. ASP	Channel & context SP
Embedding dim	192	256

Table 3: Comparison of ECAPA-TDNN and RawNet3.

V Experimental Methods

This section details our experimental methodology, including data preparation, fine-tuning protocols, and the evaluation framework used to benchmark ECAPA-TDNN and RawNet3 under resource-constrained conditions.

1. Data Partitioning and Preprocessing

To ensure strict zero-speaker leakage across splits, both datasets are partitioned into disjoint train, validation, and test subsets:

- **TidyVoice:** The official training split is preserved for training (262,268 utterances, 3,666 speakers). The source development set (808 speakers) is partitioned 50/50 into Validation (29,720 utterances, 404 speakers) and Test (29,723 utterances, 404 speakers) using a deterministic seed (seed 42), optimizing for balanced utterance counts and language distribution with zero speaker overlap.
- **ViMD:** The training split contains 15,023 utterances from 10,291 speakers. The original test split (2,026 utterances, 1,344 speakers) is kept intact. A data audit identified two overlapping singleton speakers between validation and test (spk_73_0186 and spk_76_0219); these singleton rows were pruned from the validation set to eliminate speaker leakage, yielding a clean validation split of 1,898 utterances across 1,318 speakers without reducing the available genuine trial capacity.

Table 4 summarizes the standardized data splits across both corpora.

Dataset	Split	Utterances	Speakers	Genuine Pairs
TidyVoice	Train	262,268	3,666	—
	Validation	29,720	404	7,954
	Test	29,723	404	7,898
ViMD	Train	15,023	10,291	7,044
	Validation	1,898	1,318	863
	Test	2,026	1,344	1,042

Table 4: Summary of standardized dataset splits and trial capacities.

Audio Preprocessing Pipeline. All input audio files are standardized through a unified pre-processing pipeline:

1. Decode raw audio to 32-bit floating-point arrays.
2. Downmix multi-channel audio to mono via arithmetic averaging across channels.

3. Resample audio to 16 kHz using polyphase filtering.
4. Retain native acoustic dynamic range without amplitude normalization.
5. Deterministically crop utterances: for training, audio is cropped to 48,000 samples (3.00 s) for ECAPA-TDNN and 48,240 samples (3.015 s) for RawNet3; for evaluation, segments are cropped to 48,000 samples for ECAPA-TDNN and 64,240 samples (4.015 s) for RawNet3.
6. Utterances shorter than the target segment duration are extended via sample repetition rather than zero-padding to prevent introducing synthetic silence artifacts.

2. Fine-Tuning Configuration

Both models are initialized from publicly released, state-of-the-art checkpoints pretrained on VoxCeleb:

- **ECAPA-TDNN:** Loaded from `speechbrain/spkrec-ecapa-voxceleb` (revision `0f99f2d`), comprising 20.77M encoder parameters and outputting 192-dimensional embeddings.
- **RawNet3:** Loaded from `jungjee/RawNet3` (revision `c89102e`), comprising 16.28M encoder parameters and outputting 256-dimensional embeddings.

For target-domain adaptation, the original pretrained classification heads are discarded and replaced with freshly initialized Additive Angular Margin Softmax (AAM-Softmax) [6] heads, with output dimensions matching the number of training speakers in each respective dataset (3,666 for TidyVoice, 10,291 for ViMD). During downstream verification and identification inference, the classification heads are removed and only the L2-normalized encoder embeddings are utilized.

Configuration	ECAPA-TDNN	RawNet3
Random seed	42	42
Objective function	AAM-Softmax	AAM-Softmax
AAM margin (m) / scale (s)	0.2 / 30	0.2 / 30
Precision	FP16 (scale: 1024)	FP16 (scale: 1024)
Batch size	24	24
Optimizer	Adam	Adam
Encoder learning rate	1×10^{-4}	1×10^{-4}
Head learning rate	1×10^{-4}	1×10^{-4}
Weight decay	2×10^{-6}	5×10^{-5}
Gradient clipping	Global norm 5.0	Global norm 5.0
Max training epochs	3	3
Early stopping metric	Validation EER	Validation EER
Early stopping patience	1 epoch (tie: minDCF)	1 epoch (tie: minDCF)

Table 5: Hyperparameter settings and training configurations.

Resource-Constrained Protocol. To complete all experimental runs within an 8-hour computational budget on an NVIDIA Tesla T4 GPU, training is conducted with one deterministic

utterance per speaker per epoch (3,666 examples/epoch for TidyVoice, 10,291 examples/epoch for ViMD). Batch size is fixed at 24 following memory stress-testing against the 10,291-class ViMD projection layer, preserving a 20% safety margin below GPU VRAM limits.

3. Evaluation Protocol and Metrics

Verification performance is evaluated on frozen trial lists identical across all models to ensure an unbiased comparison:

- **Trial Generation:** To prevent speakers with large utterance counts from dominating the evaluation, genuine pairs are capped at a maximum of 20 pairs per speaker. Each evaluation split is paired with 100,000 randomly sampled impostor trials, providing statistical resolution down to a False Acceptance Rate (FAR) of 0.01%.
- **Scoring:** The trial score between an enrollment embedding e_1 and a test embedding e_2 is computed as the cosine similarity:

$$S(e_1, e_2) = \frac{e_1 \cdot e_2}{\|e_1\|_2 \|e_2\|_2} \quad (1)$$

- **Primary Metrics:** We report the Equal Error Rate (**EER** in %), where False Acceptance Rate equals False Rejection Rate ($FAR = FRR$), and the minimum Detection Cost Function (**minDCF**) parameterized with $P_{\text{target}} = 0.01$ and $C_{\text{fa}} = C_{\text{miss}} = 1.0$:

$$C_{\text{Det}} = C_{\text{miss}} \cdot P_{\text{target}} \cdot FRR + C_{\text{fa}} \cdot (1 - P_{\text{target}}) \cdot FAR \quad (2)$$

- **Security Operating Point:** A security-critical threshold is calibrated on the Validation set at $FAR = 0.1\%$ and frozen. This threshold is then evaluated exactly once on the Test set to report operating accuracy, FAR, FRR, and True Acceptance Rate ($TAR = 1 - FRR$). Additionally, we compute $TAR@FAR$ across $\{5\%, 1\%, 0.1\%, 0.01\%\}$.

VI Results and Analysis

This section presents the empirical results of benchmarking ECAPA-TDNN and RawNet3 across TidyVoice and ViMD, detailing training dynamics, verification accuracy, and computational efficiency.

1. Training Dynamics and Convergence

Table 6 records the training progression across the four model–dataset pairs. Each run operated under an early stopping patience of 1 epoch on validation EER (with minDCF as tie-breaker).

Model / Dataset	Epoch	Train Loss	Val EER (%)	Val minDCF	Outcome
ECAPA-TDNN / TidyVoice	0	16.4067	3.943	0.3685	Best Checkpoint
	1	15.7529	4.075	0.4068	Early Stopped
ECAPA-TDNN / ViMD	0	17.0806	3.936	0.4503	Best Checkpoint
	1	15.3398	6.634	0.7661	Early Stopped
RawNet3 / TidyVoice	0	15.7133	5.860	0.5251	Best Checkpoint
	1	14.7825	9.693	0.7767	Early Stopped
RawNet3 / ViMD	0	15.9177	29.664	0.9164	Initialized
	1	14.8658	7.532	0.6805	Improved
	2	13.3438	2.692	0.2446	Best Checkpoint

Table 6: Training progression and checkpoint selection across four experimental runs.

A notable observation is that decreasing training loss did not consistently translate into lower validation error. For ECAPA-TDNN on both datasets and RawNet3 on TidyVoice, the models achieved their best validation metrics at Epoch 0; subsequent epochs caused validation degradation, triggering early stopping. In contrast, RawNet3 on ViMD exhibited continuous and substantial convergence, reducing validation EER from 29.66% at Epoch 0 to 2.69% at Epoch 2.

2. Speaker Verification Benchmark

Table 7 reports the primary evaluation metrics on the unseen Test splits using the best checkpoints selected on Validation.

Model	Dataset	EER (%) ↓	minDCF ↓	Accuracy (%)	FAR (%)	FRR (%)	TAR (%) ↑
ECAPA-TDNN	TidyVoice	3.912	0.3586	97.865	0.080	28.159	71.841
RawNet3	TidyVoice	5.875	0.5624	96.509	0.124	46.126	53.874
ECAPA-TDNN	ViMD	3.804	0.4516	99.527	0.116	34.741	65.259
RawNet3	ViMD	3.071	0.2262	99.765	0.075	15.547	84.453

Table 7: Final test performance on TidyVoice and ViMD. Thresholds are calibrated at FAR = 0.1% on the Validation split and evaluated once on the Test split.

Analysis of Operating Performance. High overall accuracy values ($> 96\%$) reflect the protocol’s 100,000 impostor trials relative to genuine trials. Therefore, EER, minDCF, and True Acceptance Rate (TAR) at low False Acceptance Rates serve as the discriminative metrics:

- **On TidyVoice (Multilingual):** ECAPA-TDNN achieves superior performance across all metrics, reducing EER by 33.4% relative (3.912% vs. 5.875%) and minDCF by 36.2% relative (0.3586 vs. 0.5624), while yielding a +17.97% higher TAR at the frozen security threshold.
- **On ViMD (Vietnamese Multi-Dialect):** RawNet3 significantly outperforms ECAPA-TDNN, achieving a 19.3% relative reduction in EER (3.071% vs. 3.804%) and a 49.9% relative reduction in minDCF (0.2262 vs. 0.4516). At the operational threshold, RawNet3 boosts TAR to 84.453% (compared to 65.259% for ECAPA-TDNN) while maintaining a strict 0.075% FAR.

Table 8 provides a fine-grained evaluation of authentication security, detailing the True Acceptance Rate at varying target False Acceptance Rates (TAR@FAR).

Model	Dataset	TAR@FAR 5%	TAR@FAR 1%	TAR@FAR 0.1%	TAR@FAR 0.01%
ECAPA-TDNN	TidyVoice	96.771	90.377	73.563	48.974
RawNet3	TidyVoice	93.391	82.907	48.088	0.886
ECAPA-TDNN	ViMD	97.505	88.484	63.820	37.812
RawNet3	ViMD	97.217	95.106	86.756	54.798

Table 8: True Acceptance Rate (%) at specific False Acceptance Rate thresholds (TAR@FAR).

3. Computational Efficiency and Model Selection

To evaluate suitability for real-time deployment in our virtual assistant pipeline, standalone model inference latency was benchmarked on an NVIDIA Tesla T4 GPU using CUDA events across 50 iterations with batch size 1 after 10 warm-up runs (Table 9).

Model	Parameters	Median Latency (ms)	p95 Latency (ms)
ECAPA-TDNN	20.77 M	12.023	12.912
RawNet3	16.28 M	8.840	12.884

Table 9: Model complexity and isolated GPU inference latency (batch size 1 on NVIDIA Tesla T4).

Deployment Selection Rationale. Based on our empirical findings, ****RawNet3** fine-tuned on ViMD (Epoch 2) ****** is selected as the verification engine for the Who-Speak-AI virtual assistant:

1. **Target-Domain Alignment:** The intended application is a Vietnamese-language conversational assistant. RawNet3 demonstrates clear superiority in modeling regional Vietnamese acoustic and dialectal nuances, obtaining 3.071% EER and 0.2262 minDCF.
2. **High Security Margin:** At the operational security threshold ($FAR \leq 0.1\%$), RawNet3 delivers an 84.45% True Acceptance Rate, significantly reducing user friction (lower False Rejection) during voice authentication compared to ECAPA-TDNN (65.26%).
3. **Efficiency Profile:** RawNet3 contains 21.6% fewer parameters (16.28M vs. 20.77M) and exhibits 26.5% lower median GPU latency (8.84 ms vs. 12.02 ms), preserving critical compute headroom for concurrent ASR, LLM reasoning, and TTS synthesis.

VII Application: Who-Speak-AI

1. Introduction to Who-Speak-AI

Who-Speak-AI is a voice-driven virtual assistant that integrates biometric speaker recognition into a modular conversational pipeline. By coupling speech processing with our fine-tuned RawNet3 speaker embedding model, the assistant provides seamless voice interaction while enforcing biometric verification for sensitive operations and personalizing responses based on speaker identity.

1.1 Functional Requirements

1.2 Non-functional Requirements

2. Architectural Design

3. User Interface

something something...

3.1 General Features

3.2 Secured Features

3.3 Registration

4. Continuous Integration & Continuous Delivery

5. Validity of Enrollment & Verification

6. Performance Studies

VIII Conclusion

This report benchmarked ECAPA-TDNN and RawNet3 across multilingual (TidyVoice) and Vietnamese multi-dialect (ViMD) conditions. While ECAPA-TDNN demonstrated stronger multilingual generalization (3.912% vs. 5.875% EER on TidyVoice), RawNet3 excelled on regional Vietnamese speech (3.071% vs. 3.804% EER and 0.2262 vs. 0.4516 minDCF on ViMD), accompanied by 21.6% fewer parameters and 26.5% lower GPU latency. Consequently, **RawNet3 fine-tuned on ViMD** is selected as the biometric verification engine for the *Who-Speak-AI* virtual assistant.

Future work includes unconstrained multi-epoch training with data augmentation and optimizing on-device CPU execution.

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